

Programme: Diploma in Cloud Computing and Big Data	
Course Code: 5279B	Course Title: Advanced Data Structures Lab
Semester: 5	Credits: 1.5
Course Category: Programme Elective Course	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- Develop Python programs to perform advanced operations on graphs and trees.
- Implement Maps, Hash Tables and Set data structures in Python
- Develop Python programs for text processing applications

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Sound knowledge in Python Programming	3279	Programming in Python Lab	3
Basic theoretical and programming concepts of Data Structures	4272	Data Structures and Algorithms	4
	4278	Data Structures and Algorithms Lab	4

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate graph algorithms for transitive closure, shortest path and minimum spanning tree	12	Apply
CO2	Develop programs to perform operations on AVL trees	9	Apply

CO3	Implement Maps, Hash Tables and Set data structures in Python	9	Apply
CO4	Use text processing algorithms in python	11	Apply
	Lab Test	4	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	2	2			
CO2	3	2	2	2			
CO3	3	2	2	2			
CO4	3	2	2	2		2	1

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Demonstrate graph algorithms for transitive closure, shortest path and minimum spanning tree.		
M1.01	Implement Floyd Warshall algorithm to find transitive closure of a directed graph using Python	3	Applying
M1.02	Demonstrate Dijkstra's shortest path algorithm	3	Applying
M1.03	Apply Prim-Jarnik Algorithm to find Minimum Spanning Tree of a graph	3	Applying
M1.04	Find the Minimum Spanning Tree of a graph using Kruskal's Algorithm.	3	Applying
CO2	Develop programs to perform operations on AVL trees		
M2.01	Develop Python program to perform insert operation on AVL Trees.	5	Applying

M2.02	Implement delete operation on AVL Trees.	4	Applying
	Lab Test – I	2	
CO3	Implement Maps, Hash Tables and Set data structures in Python		
M3.01	Using maps, develop a Python program for counting word frequencies in a document	2	Applying
M3.02	Implement a Hash table with separate chaining for collision resolution	2	Applying
M3.03	Implement a Hash table with linear probing for collision resolution	2	Applying
M3.04	Develop Python programs to implement the set ADT and perform set operations.	3	Applying
CO4	Use text processing algorithms in python		
M4.01	Implement pattern matching algorithm using the brute force approach	3	Applying
M4.02	Implement Huffman coding algorithm for text compression.	3	Applying
M4.03	Open ended experiments	5	Applying
	Lab Test – II	2	

Text / Reference

T/R	Book Title/Author
T1	Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser, Data Structures and Algorithms in Python (An Indian Adaptation), Wiley; 1st edition (1 July 2021)
R1	Bradley N Miller, Problem Solving with Algorithms and Data Structures Using Python, Franklin Beedle & Assoc; Second edition (1 January 2013)

R2	Kent D. Lee, Steve Hubbard, Data Structures and Algorithms with Python, Springer, 2015
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Online Resources

Sl.No	Website Link
1	https://www.edureka.co/blog/data-structures-in-python/
2	https://realpython.com/python-data-structures/
3	https://www.programiz.com/dsa